In a vector space take a set of vectors $\{v_1, v_2, \ldots, v_r\}$
The set of linear combinations of these vectors, i.e.

$$\{c_1 v_1 + c_2 v_2 + \ldots + c_r v_r\}$$

where the c_i's are any scalars, is always a subspace.
As we learned earlier, we call this the subspace
<u>spanned</u> by this set of vectors.

Why is this a subspace? Well without giving a formal
proof, let's run through the general idea. First off
0 vector is in this set via taking all c_i's to be 0. And
sum of two vectors of this type, we see that we have

$$c_1 v_1 + d_1 v_1 = (c_1 + d_1) v_1$$

still a scalar multiple of v_1 (and can do this with
each v_j). So that roughly shows why it's closed under
addition. Closed under scalar multiplication is similar.

There are many important subspaces we wish to study, that's what next section is about.

But first, we must realize that proving something is a vector space or perhaps proving a subset is a subspace is important to be able to do. After all if you are going to use a property of subspace on a specific subset, you have to be able to prove it's a subspace!

To that end, let us prove a couple "simple" results about vector spaces. I say simple in quotes because these will look very easy, but proving them might not be! In general you can't assume because something is "obvious" in $\mathbb{R}$ that it is in any vector space, might not even be true in every vector space!

In any vector space V we have

$$\underbrace{0}_{\text{scalar}} \cdot \underbrace{u}_{\text{any vector}} = \underbrace{0}_{\text{vector}}$$

In words: the 0 scalar times any vector gives the 0 vector.

This is definitely a result which appears to be "obviously" true based on the way it looks. But it's not one of the given properties of vector spaces so how do you know it's true?

Have to be careful about jumping to conclusions, after all if $ab = 0$ does that imply $a = 0$ or $b = 0$? Well if a and b are real numbers yes. If they are matrices, it doesn't!

So have to be careful in general about claiming something is true just because it seems to work in $\mathbb{R}$.

Being true in real numbers doesn't guarantee it's true in every vector space. That's part of what makes this hard, sometimes you have to go against your instincts.

So why is the statement $0 \cdot u = 0$ true in any vector space? Observe

$$0u = (0 + 0)u = 0u + 0u$$

since that's how real number scalar 0 works in first step, then using distributive law in second step. After that, since every element has additive inverse we can cancel the $0u$ to give the desired

$$0 = 0u$$

We also have

$$\underbrace{c}_{\text{any scalar}} \cdot \underbrace{0}_{\text{vector}} = \underbrace{0}_{\text{vector}}$$

I.e. the 0 vector times any scalar is still 0 vector. Why?

$$c \cdot 0 = c(0 + 0) = c \cdot 0 + c \cdot 0$$

First step from 0 vector plus any vector is same vector and then distributive law (albeit different version than prior problem). Then using additive inverse again

$$0 = c \cdot 0$$

as desired. This type of argument often needed in vector spaces, can't take things which are true in $\mathbb{R}$ for granted always as stated.

But you also see how important the distributive laws are. Without those to connect the two operations, you wouldn't be able to prove these things!

So as "obvious" as they are, if you didn't have that connection between the two operations, these would not hold!

So if you are asked to prove something is a vector space, distributive laws are something you have to verify to work, again do not just assume they work because they look natural and are known to work in $\mathbb{R}$! In other structures they might not work.

4.2 Specific Subspaces

Now that we have defined subspaces, we want to talk about some specific ones which are important to us.

Fortunately we start in context which is reasonably familiar to us from earlier in the course, largely just formalizing the already mentioned ideas.

For an $m \times n$ real matrix A, the <u>null space</u> of A, denoted Nul A, is the set of vectors in $\mathbb{R}^n$ ($n \times 1$ so the multiplication makes sense) which A maps to 0.

I.e. all $x \in \mathbb{R}^n$ such that $Ax = 0$

As we alluded to last class, the set of such x is a subspace as $A(0) = 0$ always works so the 0 vector belongs as it needs to. If x, y are in this set, then

$$0 = Ax + Ay = A(x + y)$$

showing $x + y$ belongs as well (closed under addition as also needed). And scalar multiplication similar so this is a subspace.

As also alluded, for some nonzero b, solutions to $Ax = b$ won't be subspace, right away 0 vector is not in the set.

So how to find null space?

4.2.5

$$A = \begin{bmatrix} 1 & -2 & 0 & 4 & 0 \\ 0 & 0 & 1 & -9 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

Since this matrix is already in REF (RREF in fact), we note that the second and fourth columns are the nonpivot columns/correspond to free variables. The three equations are:

$$x_5 = 0$$
$$x_3 - 9x_4 = 0$$
$$x_1 - 2x_2 + 4x_4 = 0$$

So expressing the three basic variables, x_1, x_3, x_5 in terms of the two free variables, x_2, x_4 yields

$$x_5 = 0 \quad , \quad x_3 = 9x_4 \quad , \quad x_1 = 2x_2 - 4x_4$$

So general solution is

$$\begin{bmatrix} 2x_2 - 4x_4 \\ x_2 \\ 9x_4 \\ x_4 \\ 0 \end{bmatrix}$$

which in parametric vector form is

$$x_2 \begin{bmatrix} 2 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} + x_4 \begin{bmatrix} -4 \\ 0 \\ 9 \\ 1 \\ 0 \end{bmatrix}$$

The major thing is that this subspace is spanned by

$$\left\{ \begin{bmatrix} 2 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} -4 \\ 0 \\ 9 \\ 1 \\ 0 \end{bmatrix} \right\}$$

i.e., that is a <u>spanning set</u> for that subspace. That is the key way to describe a subspace, to give a spanning set.

Again subspaces usually have infinitely many vectors, can't list them one by one. Giving a spanning set is a way to compactly describe the subspace.

This was in a sense the major reason why we wanted parametric vector form for solution set, it is the most convenient form for determining a spanning set.

So to find null space you do generally need to put matrix in REF, can naturally take some time.

Another subspace of interest to us is the space generated by the columns of A, denoted Col A.

We were often curious if these columns spanned the full space $\mathbb{R}^m$ (this was determined by whether or not A had a pivot in every row).

But when they do not span the full space, what do they span? Well a subspace, namely the column space.

And fortunately for that, there is a very apparent spanning set, namely the columns themselves.

So column space is much easier to find than null space, but a natural question is: do we need all the columns to generate the column space or might some be redundant? Definitely a question we will want to more carefully address later on.

As we learned earlier, another way to think about what vectors are linear combinations of the columns is asking for which b the matrix equation $Ax = b$ has a solution for.

So given that Nul A has to with solving $Ax = 0$ and Col A has to do with solving $Ax = b$, are they comparable to each other given the former seems to do with solving the homogeneous equation while the latter with solving a nonhomogeneous equation?

They really aren't directly comparable. Big issue is Nul A is subspace of $\mathbb{R}^n$ while Col A is subspace of $\mathbb{R}^m$. So they generally aren't even in same space!

Nul A has to do with finding x so that $Ax = 0$. Col A has to do with finding b for which $Ax = b$ has solution. So Nul A is looking for x, Col A looking for b, so not really at all same thing even if the two equations involved seem related.

For again A an $m \times n$ real matrix, we also have the <u>row space</u> of A, denoted Row A, where you take the span of the rows of A as vectors in $\mathbb{R}^n$.

This is really no different in idea than the columns, just we were apt to use columns as vectors for things like solving linear systems since notationally, the way we did the matrix multiplication, it meshed that way.

But again nothing inherently different in thinking of the of rows as vectors as well (of course it's a different set of vectors than the columns).

Row A and Nul A are both subspaces of the same space $\mathbb{R}^n$, not like Col A which is subspace of $\mathbb{R}^m$. But even so they aren't really comparable still, usually completely different subspaces. The actual relationship between those becomes quite important later in the class (Chapter 6).

When talking about a matrix's "vectors" do we mean its rows or columns or something else even? Sure mostly talked about its columns prior, but that could mean, many things in broader contexts.

That is why it is inappropriate to ask something like "Is this matrix linearly independent?". Sure based on recent things you would think that is talking about the matrix's columns, but in grand scheme of things can't really assume that, could be rows.

Hence that question is ambiguous, you would really need to ask "Are the columns of this matrix linearly independent?"

And sometimes rows might be linearly dependent and columns linearly independent so that can certainly make a difference (in nonsquare matrices).

For a linear transformation $T: \mathbb{R}^n \dashrightarrow \mathbb{R}^m$, the <u>kernel</u> of T is the set of vectors that T maps to 0, which is also the null space of the matrix associated with T.

Term kernel is used alot in higher math though maybe not a ton in this class.

It's important because in many contexts we are interested in what points a function maps to 0. But if we are in a context where the function can't be represented by a matrix, we can't really use the term null space so the term kernel is more versatile there.

Another interesting note is that the range of such a T is the column space of T's associated matrix. That's perhaps simplest way to express the range.

4.2.9

$$\left\{ \begin{bmatrix} a \\ b \\ c \\ d \end{bmatrix} : \begin{array}{l} a - 2b = 4d \\ 2a = c + 3d \end{array} \right\}$$

The set of points satisfying this is a subset of $\mathbb{R}^4$ (quite clearly vector has four components). But it's definitely not the full space as not every point in $\mathbb{R}^4$ satisfies these equations. But is this subset a subspace? Again those terms don't mean the same thing - do not confuse them!

This is a very common style question, to ask if a subset is a subspace. But how to answer? Well default way would be to check the three properties needed to make a subset a subspace: contains 0 vector, closed under addition, and closed under scalar multiplication.

Can we do that here? Yes, just maybe a little annoying notationally for latter two, perhaps there's an easier way?

These two linear equations required, think of the matrix equation which models them. That will be:

$$\begin{bmatrix} 1 & -2 & -4 & 0 \\ 2 & 0 & -1 & -3 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \\ d \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

Again this gives the two exact equations governing our set of points.

Therefore that set of points is the null space of this matrix on the far left. That being so, it is a subspace since null space is always a subspace.

So we sort of indirectly established this is subspace, kind of backdoored it, didn't check the individual properties needed for subspace as we might have to in other circumstances.

This is a reason why Nul A, Col A, Row A can be useful.

Gave a worksheet question recently where I gave you the solution set and asked you to "recover" the system. Was really preluding to this concept with that question.

4.2.11

$$\left\{ \begin{bmatrix} b - 2d \\ 5 + d \\ b + 2d \\ d \end{bmatrix} : b, d \text{ real} \right\}$$

Again is this a subspace of $\mathbb{R}^4$?

Well can't have $5 + d$ and d equal to 0 at same time so can't have the second and fourth components both be 0 so we can't create the 0 vector here so this cannot be a subspace.

The +5 in one component is sort of a tipoff to that as that makes this feel nonhomogeneous in a sense.

Indeed if we were going to try to show this is a subspace by writing it as a span we would get to

$$b \begin{bmatrix} 1 \\ 0 \\ 1 \\ 0 \end{bmatrix} + d \begin{bmatrix} -2 \\ 1 \\ 3 \\ 1 \end{bmatrix} + \begin{bmatrix} 0 \\ 5 \\ 0 \\ 0 \end{bmatrix}$$

If it was just first two, would be subspace, span of those. The latter one fouls this up, the same type of extra term that appears in the general solution to a nonhomogeneous linear system.

4.3 Basis

How do we generally describe a subspace? By giving a spanning set.

Ideally we'd like to give the most efficient spanning set though, no redundancy.

We stated that finding column space is easier than null space since no row operations needed. But we also worried that we might not need all of the columns, that it could be redundant. Didn't have that concern about null space.

Say we had a 5×100 matrix, surely we wouldn't need all 100 columns to create the subspace in $\mathbb{R}^5$ that those span?

In $\mathbb{R}^3$, the vectors e_1, e_2, e_3 are a spanning set. After all, they form identity matrix when written as columns of matrix which is certainly invertible.

Say we took five vectors which spanned $\mathbb{R}^3$ instead, perhaps the five columns in 4.2.5

Those span $\mathbb{R}^3$ but do we need all of them? Since we can do it with three, feels like we shouldn't. But which ones are redundant if so and how many? This is the major topic that 4.3 clarifies.